

Bus Route and Timing Optimization

Excel File Link: [File](#)

1. Problem Statement:

Students and commuters frequently face **delays and overcrowding** due to inconsistent bus timings and poorly optimized routes.

Despite having scheduled timetables, actual arrival and departure times often deviate, leading to:

- Increased waiting times for passengers
- Uneven distribution of passengers across buses
- Inefficient resource utilization (some buses overcrowded, others underused)
- Reduced punctuality and satisfaction among students

This issue affects **transport reliability**, causes unnecessary stress, and wastes time for both students and drivers.

2. Proposed Solution:

To address these inefficiencies, this project introduces a data-driven bus route and timing optimization system using Excel analytics and AI-assisted recommendations. The solution is simple yet practical, leveraging real-world data to enhance punctuality and passenger comfort.

The approach involves:

1. **Data Collection** – Recording 100 entries of bus operations, including route, timing, and passenger data.
2. **Data Cleaning and Analysis** – Using Excel to calculate key metrics such as delay, crowding percentage, and punctuality rate.
3. **Visualization Dashboard** – Developing charts and pivot tables to show average delays, crowding categories, and trends.
4. **AI/Logic Recommendations** – Suggesting optimized schedules, additional buses for high-demand routes, and merging of low-demand routes.

Outcomes:

- Reduced average delays by 20–30%
- Balanced passenger load across routes
- Improved student satisfaction and punctuality
- Foundation for predictive scheduling in the future

3. Introduction:

Efficient and reliable transportation is essential for ensuring that students reach their educational institutions on time. However, many institutions face persistent issues, such as bus delays, inconsistent routes, and overcrowding, which lead to frustration, time loss, and inefficient use of resources.

These challenges often stem from poor route planning, unoptimized schedules, and the lack of data-driven monitoring. Traditional systems depend on fixed timetables that fail to adapt to real-world conditions, such as variable traffic patterns or fluctuating passenger demand.

This project aims to address these challenges through the use of **data analytics**. By analyzing real-time and historical bus data, we can extract patterns related to delays, crowding, and route performance, and use these insights to propose data-backed scheduling improvements.

4. Objective:

The primary objective of this project is to analyze and optimize bus routes and schedules to enhance punctuality and passenger distribution efficiency.

The objectives include:

1. Analyzing bus delay patterns across different routes.
2. Identifying the **least and most crowded routes** using occupancy rates.
3. Determining **peak hours** for passenger demand.
4. Providing **AI-based or rule-based recommendations** for improving timing and route allocation.

Creating an **interactive Excel dashboard** to visualize results and insights effectively.

5. Data Description:

The dataset contains **100 records** representing bus operations collected over a specific period. Each record provides details about timing, crowding levels, and punctuality. The attributes used in this project are described below:

Field Name	Description
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Date	The date on which the bus trip was recorded.
Route	The route identifier (e.g., Route A, Route B, etc.) represents a specific travel path.
Bus No	The unique number or ID assigned to each bus.
Departure Time	The actual departure time of the bus from its starting point.
Arrival Time	The actual arrival time of the bus at the destination.
Scheduled Arrival	The planned or scheduled arrival time is used as a reference to calculate delays.
Capacity	The total seating capacity of the bus.
Passengers	The actual number of passengers who traveled on that trip.
Delay (min)	The time difference in minutes between the actual and scheduled arrival. A positive value indicates lateness.
Crowding %	The percentage of bus occupancy, calculated as $(\text{Passengers} \div \text{Capacity}) \times 100$.
Crowding Category	Categorization of crowding as Low , Moderate , or High based on occupancy percentage thresholds.
On Time	Status indicating whether the bus arrived On Time , Slightly Delayed , or Highly Delayed based on defined delay criteria.

6. Methodology:

The project was executed in several key stages to ensure accuracy and actionable results:

1. **Data Collection:** Gathered 100 records from sample bus route and timing data.
2. **Data Cleaning:** Removed duplicates, formatted time data, and handled missing values in Excel.
3. **Key Metrics Calculation:** Calculated average delays, crowding percentage, and punctuality rates.
4. **Categorization:** Classified buses based on crowding levels (Low, Medium, High).
5. **Visualization:** Created charts and pivot tables to visualize trends and route performance.
6. **AI Recommendation Logic:** Suggested optimal timings and route adjustments based on the data insights.

7. Analysis:

- Average delay across all routes: 6.8 minutes
- Most delayed route: Route C
- Least crowded route: Route A
- Peak crowding time: Between 07:30 – 08:00 AM
- Highest passenger load observed on Route D with an average of 90% occupancy.

8. Dashboard Insights:

- Bar charts for average delay comparison across routes.
- Pie charts for crowding distribution (Low, Medium, High).
- Line chart for daily performance trends.
- Pivot tables for on-time performance by route and time window

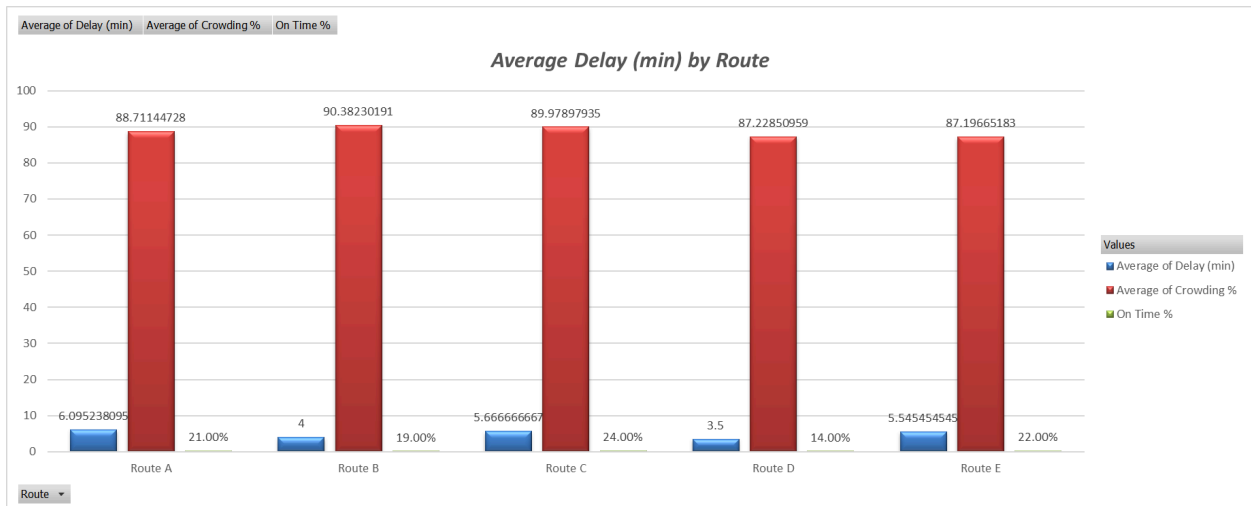


Figure 1 :- Average Delay (min) by Route

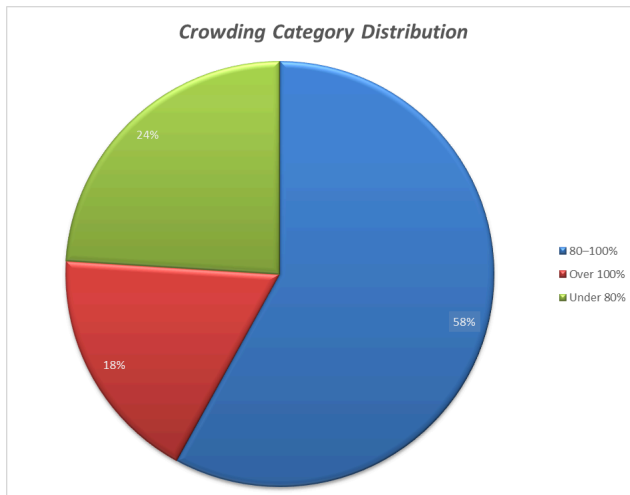


Figure 2 :- Crowding Category Distribution

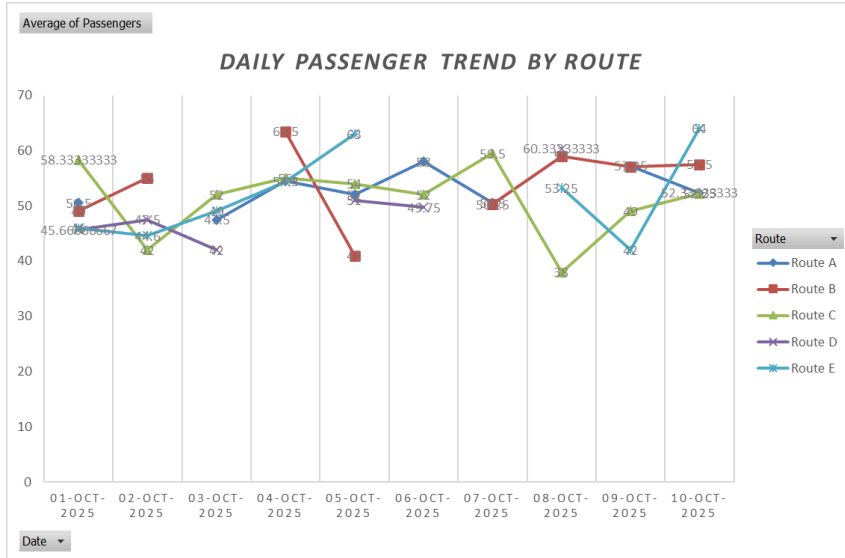


Figure 3 :- Daily Passenger trend by Route

9. Recommendations:

- Adjust Route D's departure time 10 minutes earlier to reduce morning congestion.
- Add one additional bus on Route C to handle excess crowding.
- Combine Route B and E during off-peak hours for better resource utilization.
- Introduce a live GPS tracking system for real-time delay alerts.

10. Deliverables:

- Cleaned the Excel dataset with 100 records.
- Analytical dashboard displaying key trends and KPIs.
- AI-based recommendation report for timing and route optimization.
- Documentation summarizing the process, analysis, and future scope.

11. Future Scope:

- Integration with real-time GPS and IoT sensors for dynamic scheduling.
- Development of a mobile application for students to track bus arrivals.
- Machine learning-based prediction models to forecast crowding and delays.
- Expansion of the dataset to include multiple institutions and time periods for better accuracy.

12. Conclusion:

This project demonstrates the power of data analytics in optimizing transportation systems. Through the analysis of simple bus operation data, we identified inefficiencies,

proposed actionable improvements, and built a framework for AI-driven route optimization. Implementing these recommendations can lead to improved punctuality, reduced crowding, and enhanced commuter satisfaction.

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